Personalizing motor control in predictive gait simulations of children with cerebral palsy using muscle synergies

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Introduction

Predictive gait simulations have the potential to investigate how patient-specific impairments contribute to altered walking in children with cerebral palsy (CP). While musculoskeletal impairments such as muscle weakness and contractures are commonly incorporated into personalized models, motor control is generally less individualized [1]. Muscle synergies provide a low-dimensional representation of muscle coordination and have been used to characterize impaired motor control in CP [2]. This study investigated how incorporating EMG-derived muscle synergy information affects predicted gait kinematics and muscle activity in children with CP.

Methods

Predictive gait simulations were generated for five children with CP using PredSim [3]. Subject-specific muscle synergies were extracted bilaterally from experimental EMG recordings of eight lower-limb muscles using non-negative matrix factorization.

Three simulation configurations were compared: a subject-scaled musculoskeletal model (Scaled), a model in which muscle control was restricted to the experimentally determined number of synergies (NSyn), and a model additionally constrained by the subject-specific muscle weights within each synergy (SynW).

Predicted sagittal hip, knee, and ankle angles were compared with experimental kinematics. Simulated muscle activations were compared with EMG recordings to assess whether increasing motor control personalization improved agreement with experimental gait.

Results

Adding subject-specific synergy constraints did not consistently improve agreement with experimental gait. Mean joint-angle RMSE was lowest with the Scaled model at the hip (12.47°), knee (13.37°), and ankle (11.38°), compared with NSyn (12.63°, 16.66°, and 12.33°) and SynW (16.74°, 16.76°, and 15.18°), respectively. Muscle activity showed a similar overall trend, with mean correlation coefficients of 0.28, 0.19, and 0.18 for Scaled, NSyn, and SynW, respectively.

Figure 1 shows experimental and predicted sagittal kinematics and selected muscle activation periods for the right side of CP3.

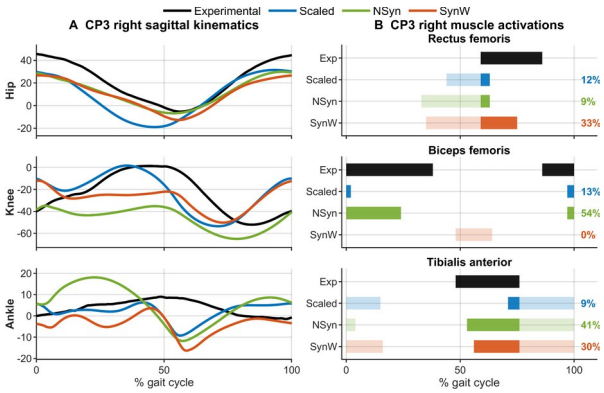


Figure 1: Experimental and predicted sagittal kinematics and selected muscle activation periods for CP3 right side. Light bars show predicted activation periods and darker segments indicate overlap with experimental EMG.

Subject	Scaled	NSyn	SynW
CP 1	8.64	8.41	18.12
CP 2	12.36	10.94	12.39
CP 3	11.78	15.75	12.21
CP 4	16.51	10.64	15.93
CP 5	12.74	23.62	21.49

Table 1: Mean joint-angle RMSE [deg] for each participant and simulation condition. Values were averaged across sagittal hip, knee and ankle angles from both sides.

Conclusions

Including subject-specific muscle synergy information in predictive simulations did not consistently improve the prediction of gait kinematics or muscle activity in this cohort. The results show that incorporating motor control information through synergy constraints is possible, but the current implementation may impose overly restrictive control in some cases. Further work should investigate how these constraints are formulated and weighted to better represent patient-specific motor control.

References

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3. D'Hondt et al., *10th IEEE RAS/EMBS BioRob* , 2024.